

LINEAR EQUATIONS IN TWO VARIABLES

4.1 Introduction

You have studied linear equations in one variable in the previous classes. Recall that a linear equation in one variable is an equation involving only linear polynomials in one variable. Thus, equations of the form $ax = b$, $ax + b = c$, $ax + b = cx + d$ (where $a \neq 0$, b, c, d are constants) are linear equations in one variable. For example, $x + 2 = 0$, $2x - \sqrt{3} = 0$, $5x + 3 = 3x - 5$, $\frac{y}{2} - 3 = \frac{y}{3} + 3$ etc. are linear equations in one variable. You also know that such equations have a unique (i.e. one and only one) solution. Now, let us see some examples of solving linear equations in one variable (i.e. finding the value of the variable satisfying the equation).

Example 1 : Solve $4x + 3 = 15$

Solution :

$$4x + 3 = 15$$

$$\text{or } 4x = 15 - 3 \quad (\text{by transposition})$$

$$\text{or } 4x = 12$$

$$\therefore x = \frac{12}{4} = 3.$$

Example 2 : Solve $4x + 3 = 2x + 5$

Solution :

$$4x + 3 = 2x + 5$$

$$\text{or } 4x - 2x = 5 - 3 \quad (\text{by transposition})$$

$$\text{or } 2x = 2$$

$$\therefore x = 1.$$

Example 3 : Solve $\frac{y}{3} + 1 = \frac{y}{4} - 2$

Solution :

$$\frac{y}{3} + 1 = \frac{y}{4} - 2$$

$$\text{or } \frac{y}{3} - \frac{y}{4} = -2 - 1 \quad (\text{by transposition})$$

$$\text{or } \frac{y}{12} = -3$$

$$\therefore y = -3 \times 12 = -36.$$

4.2 Linear Equations in Two Variables

Let us now consider the statement "The sum of two numbers is 10". If we denote the first number by x and the second number by y , then the statement can be expressed as " $x + y = 10$ ". This is a linear equation in two variables (x and y). Likewise $3x + y = -5$, $2u - v = 4$, $-4x + 3y = 9$, $\frac{x}{3} + \frac{y}{4} = 1$ etc. are all linear equations in two variables.

Thus, a linear polynomial in two variables equated to zero gives a linear equation in two variables. It is also called a first degree equation in two variables.

The general form of a linear equation in two variables is $ax + by + c = 0$, where a , b , c are constants and a , b are both non-zeroes.

If either $a = 0$ or $b = 0$, the equation $ax + by + c = 0$ is no more a linear equation in two variables in its most general form. But in some special circumstances, either a or b is granted to be zero. For example, the equation $3x + 2 = 0$ can be written in the general form as $3x + 0.y + 2 = 0$.

4.3 Solution of a Linear Equation in Two Variables

You have already known that every linear equation in one variable has a unique solution. What can you say about the solution of a linear equation in two variables? If the two variables be x and y , then a pair of values, one for x and one for y which satisfy the equation is called a solution of the equation. Let us consider the equation $3x + y = 10$. Here, $x = 3$ and $y = 1$ constitute a solution of this equation because the equation holds when $x = 3$ and $y = 1$, for $3x + y = 3 \times 3 + 1 = 10$.

This solution is written as an ordered pair $(3, 1)$. In such an ordered pair the first component invariably represents a value of x and the second, the corresponding value of y . Similarly, the ordered pair $(2, 4)$ is also a solution of the equation $3x + y = 10$. But $(2, 5)$ is not a solution of this equation, because on putting $x = 2$ and $y = 5$, we get $3x + y = 6 + 5 = 11 \neq 10$.

You have already seen two solutions of the equation $3x + y = 10$ viz. $(3, 1)$ and $(2, 4)$. You can also verify that $(0, 10)$ is also another solution of the equation. In fact, we can get many solutions of the equation by assigning different values to x (or y) in the equation and finding the corresponding value of the other. For example, if we take $x = 1$ in the equation, we get $3 + y = 10$ i.e. $y = 7$. So, $(1, 7)$ is another solution of the equation $3x + y = 10$. Again, taking $x = -2$, we get $y = 16$ so that $(-2, 16)$ is also another solution of the equation. Thus, there is no end to different solutions of a linear equation

Similarly, taking $y = 0$, the given equation reduces to $2x = 6$, which gives $x = 3$. So, $(3, 0)$ is also a solution of the given equation.

Again, taking $x = 6$, we get $12 + 3y = 6$ i.e. $y = -2$. Therefore, $(6, -2)$ is also another solution of the equation.

Hence, three solutions of the equation $2x + 3y = 6$ can be given as $(0, 2)$, $(3, 0)$ and $(6, -2)$.

- (ii) Taking $x = 0$, the equation $3x + 4y = 0$ reduces to $4y = 0$ i.e., $y = 0$. So, $(0, 0)$ is a solution of the equation $3x + 4y = 0$.

Again, taking $x = 4$, we get $12 + 4y = 0$ i.e., $y = -3$. Therefore, $(4, -3)$ is a solution of the given equation.

Similarly, taking $y = 3$, we get $3x + 12 = 0$ i.e., $x = -4$. So, $(-4, 3)$ is also another solution of the equation.

Hence, three solutions of the given equation are $(0, 0)$, $(4, -3)$ and $(-4, 3)$.

- (iii) Writing the equation $2x + 5 = 0$ as $2x + 0.y + 5 = 0$, we obtain $x = -\frac{5}{2}$ for any value of y . Hence any three solutions of the given equation are $(-\frac{5}{2}, 0)$, $(-\frac{5}{2}, 1)$ and $(-\frac{5}{2}, 2)$.

EXERCISE 4.1

1. Solve the following linear equations in one variable :

- (i) $\frac{x}{3} = -2$ (ii) $4x + 3 = 15$ (iii) $3x + 2 = x + 4$
 (iv) $9x - 4 = 32$ (v) $3y + 5(2 - y) = -16$ (vi) $13t - 4(t + 8) - 4 = 0$
 (vii) $(x - 2)(x - 3) + 2(x - 1)(x - 3) = 3(x - 1)(x - 2)$
 (viii) $\frac{x}{2} + 5 = \frac{x}{3} + 7$ (ix) $\frac{x+2}{3} - \frac{x-1}{2} = 1$
 (x) $2(x - 15) = 5(x - 11) + 4$

2. Write linear equations in two variables to represent the following statements :

- (i) The cost of a pen is eight times that of a pencil.
 (ii) The difference of two numbers is 5.

in two variables. In other words, a linear equation in two variables has infinitely many solutions. However, an arbitrary ordered pair need not be a solution of the equation.

Example 1 : The cost of a notebook is twice that of a pen. Write a linear equation to represent this statement.

Solution :

Let the cost of a notebook be ₹ x and that of a pen be ₹ y .

Then we have,

$$x = 2y$$

$$\therefore x - 2y = 0.$$

This is the linear equation which represents the given statement.

Example 2 : Write each of the following equations in the form $ax + by + c = 0$ ($a > 0$) and indicate the values of a , b and c in each case :
(i) $3x - 4y = 5$, (ii) $x + 5 = 2y$ and (iii) $3x = y$.

Solution :

(i) The equation $3x - 4y = 5$ can be written as $3x - 4y - 5 = 0$, which is of the form $ax + by + c = 0$. Here $a = 3$, $b = -4$ and $c = -5$.

(ii) The equation $x + 5 = 2y$ can be written as $x - 2y + 5 = 0$, which is of the form $ax + by + c = 0$. Here $a = 1$, $b = -2$ and $c = 5$.

(iii) The equation $3x = y$ can be written as $3x - y + 0 = 0$, which is of the form $ax + by + c = 0$, where $a = 3$, $b = -1$ and $c = 0$.

Example 3 : Write each of the following as an equation in two variables :

(i) $4x = 3$ (ii) $x = -2$ (iii) $2y = 5$

Solution :

(i) $4x = 3$ can be written as $4x + 0.y = 3$ or $4x + 0.y - 3 = 0$.

(ii) $x = 2$ can be written as $x + 0.y - 2 = 0$.

(iii) $2y = 5$ can be written as $0.x + 2y - 5 = 0$.

Example 4 : Find any three different solutions for each of the following equations :

(i) $2x + 3y = 6$ (ii) $3x + 4y = 0$ (iii) $2x + 5 = 0$.

Solution :

(i) Taking $x = 0$, the equation $2x + 3y = 6$ reduces to $3y = 6$, which gives $y = 2$. Therefore, $(0, 2)$ is a solution of the equation $2x + 3y = 6$.

- (iii) Two tables and three chairs cost ₹ 1200.
 (iv) Two numbers are in the ratio 2 : 3.
3. Find four different solutions for each of the following equations in two variables:
- (i) $x = 3y$ (ii) $4x + 3y = 12$ (iii) $x + 2y = 6$
 (iv) $5x + 2y = 0$ (v) $3x + 4 = 0$ (vi) $2y - 5 = 0$
4. Check which of the following are solutions of the equation $3x - 2y = 6$ and which are not :
- (i) $(2, 0)$ (ii) $(0, 3)$ (iii) $(3, 0)$
 (iv) $(0, -3)$ (v) $(-2, -6)$ (vi) $(4, 3)$
5. Find the value of k if
- (i) $(1, 2)$ is a solution of $3x + 2y = k$
 (ii) $(2, -3)$ is a solution of $kx - 3y + 5 = 0$.

ANSWERS

1. (i) -6 (ii) 3 (iii) 1 (iv) 4 (v) 13
 (vi) 4 (vii) $\frac{3}{2}$ (viii) 12 (ix) 1 (x) 7
2. (i) $x - 8y = 0$ (ii) $x - y = 5$
 (iii) $2x + 3y = 12000$ (iv) $3x - 2y = 0$
3. (i) $(0, 0), (3, 1), (-6, -2), (1, \frac{1}{3})$ (ii) $(0, 4), (3, 0), (-3, 8), (1, \frac{8}{3})$
 (iii) $(0, 3), (6, 0), (2, 2), (-2, 4)$ (iv) $(0, 0), (2, -5), (-2, 5), (1, -\frac{5}{2})$
 (v) $(-\frac{4}{3}, 0), (-\frac{4}{3}, 1), (-\frac{4}{3}, 2), (-\frac{4}{3}, 3)$
 (vi) $(0, \frac{5}{2}), (1, \frac{5}{2}), (2, \frac{5}{2}), (3, \frac{5}{2})$
4. (i) Yes (ii) No (iii) No (iv) Yes (v) Yes (vi) Yes
5. (i) 7 (ii) -7

4.4 Graph of a Linear Equation in two Variables

From the previous article you know that a linear equation in two variables has infinitely many solutions and you have obtained some of these solutions algebraically so far. Now, let us look at their geometrical representation. Consider the following equation :

$$2x + 3y = 12 \quad \dots (i)$$

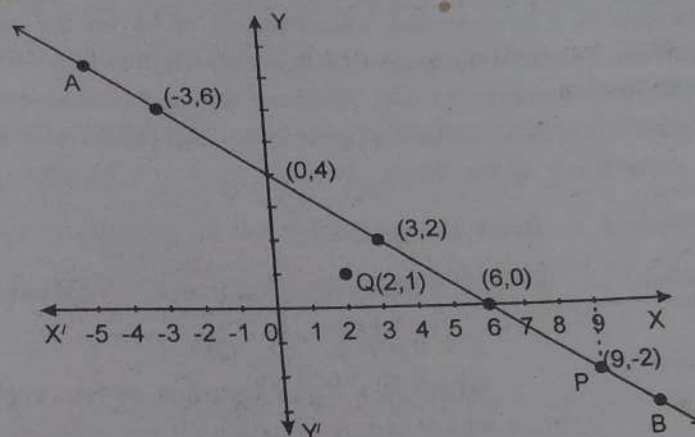


Fig. 4.1

Some of the solutions of this equation are $(0, 4)$, $(6, 0)$, $(3, 2)$, $(-3, 6)$. The points represented by these ordered pairs are plotted in the cartesian plane. Joining any two of these points we obtain a straight line $\overline{AB}$ (Fig. 4.1). Also we see that the other two points lie on the line $\overline{AB}$.

Now, let us take a point on this line, say P. On finding the co-ordinates of P, we find the ordered pair $(9, -2)$. We see that $(9, -2)$ is a solution of the equation for $x = 9$ and $y = -2$ satisfy the equation. Similarly, the co-ordinates of any other point on the straight line $\overline{AB}$ may be shown to satisfy the equation. But the co-ordinates of a point which is outside the line $\overline{AB}$ do not satisfy the equation. For example, the point $Q(2, 1)$ is not on the line $\overline{AB}$ and its co-ordinates do not satisfy equation (i) as such $(2, 1)$ is not a solution of this equation.

Thus, we observe that

- (i) every point whose co-ordinates satisfy equation (i) lies on the line $\overline{AB}$;
 - (ii) the co-ordinates of every point on the line $\overline{AB}$ constitute a solution of equation (i);
- and
- (iii) the co-ordinates of any point which does not lie on the line $\overline{AB}$ do not constitute a solution of equation (i).

So, the linear equation (i) is geometrically represented by the line $\overline{AB}$. This line is called the graph of the linear equation. Thus, we conclude that the graph of a linear equation is a straight line whose points make up the aggregate of solutions, of the equation. To obtain the graph of a linear equation in two variables, it is enough to plot two points corresponding to two solutions of the equation and draw a line through them. However, it is better to plot at least three such points so that you can immediately check the correctness of the graph.

Example 1 : Draw the graph of $x = 5$.

Solution : $x = 5$ is a linear equation in one variable. However, it can be written as

$$x + 0.y = 5 \quad \dots (i)$$

which is a linear equation in two variables. Some of the ordered pairs which satisfy equation (i) are given below :

$(5, 0), (5, 2), (5, 5), (5, -3)$ etc.

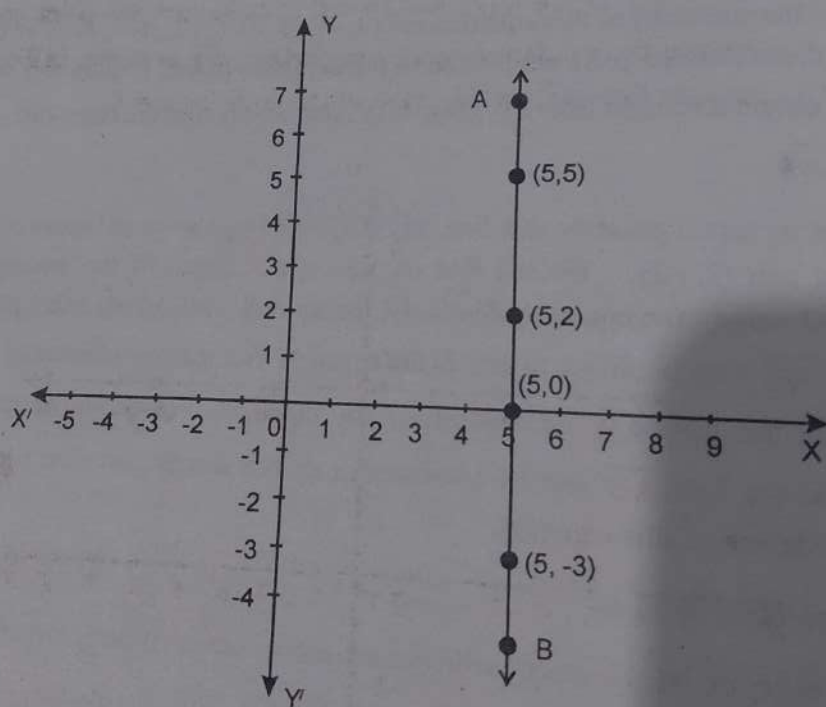


Fig. 4.2

The points represented by the above ordered pairs are plotted in the cartesian plane. Joining these points we get a straight line $\overline{AB}$ (Fig. 4.2). The line $\overline{AB}$ thus obtained is the graph of the equation $x = 5$.

- Remarks :**
- (i) The graph of the equation $x = c$, where c is a constant, is a straight line parallel to the Y-axis and cuts the X-axis at $(c, 0)$. The abscissa of any point lying on this line is c .
 - (ii) The graph of $x = 0$ is the Y-axis.

Example 2 : Draw the graph of $y = 4$.

Solution : $y = 4$ is a linear equation in one variable.

Writing this equation in the form

$$0.x + y = 4$$

we have a linear equation in two variables x and y . Some of the ordered pairs satisfying the above equation are given below :

$(0, 4), (2, 4), (5, 4), (-3, 4)$ etc.

The points represented by the above ordered pairs are plotted. Joining these points we get a straight line $\overline{AB}$ as shown in Fig. 4.3. The straight line $\overline{AB}$ is the graph of the equation $y = 4$.

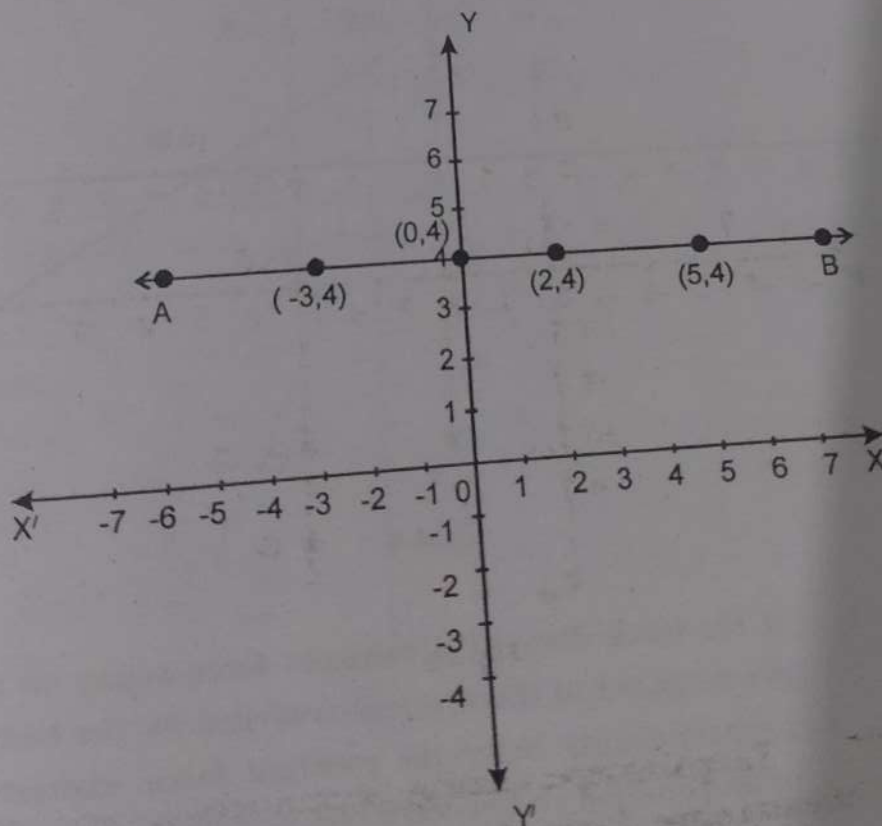


Fig. 4.3

- Remarks :** (i) The graph of the equation $y = c$, where c is a constant, is a straight line parallel to the X-axis. It cuts the Y-axis at the point $(0, c)$. The ordinate of any point on this line is c .
- (ii) The graph of $y = 0$ is the X-axis.

Example 3 : Draw the graph of $3x + 5y = 15$.

Solution : $3x + 5y = 15$ is a linear equation in two variables x and y . Some of the ordered pairs satisfying this equation are $(0, 3)$, $(5, 0)$, $(10, -3)$, $(-5, 6)$ etc. Plotting the points represented by these ordered pairs and joining them we get a straight line $\overline{AB}$ as shown in Fig. 4.4. The straight line $\overline{AB}$ is the graph of the equation $3x + 5y = 15$.

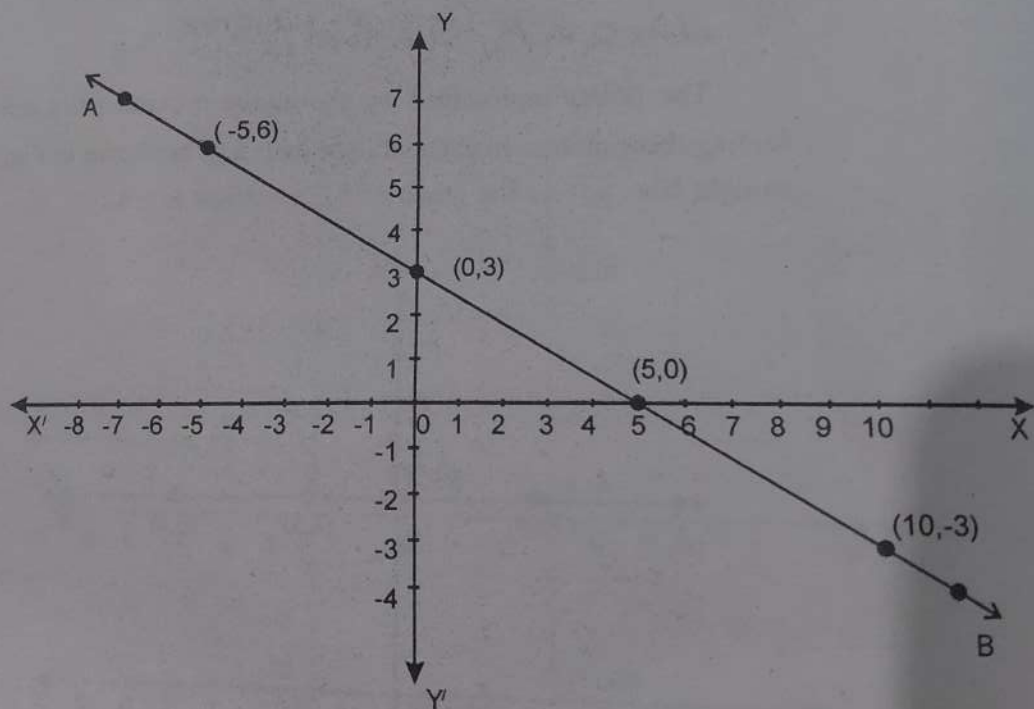


Fig. 4.4

Example 4 : If the work done by a constant force acting on a body is directly proportional to the distance travelled by the body, the constant of proportionality being the constant force, express this in the form of an equation in two variables and draw the graph of the same by taking the constant force as 2 units. Also read from the graph the work done when the distance travelled by the body is 5 units.

Solution :

Here the variables involved are distance and work. Let x be the distance travelled by the body and the work done by the force be y units. It is given that $y \propto x$. If k be the constant force acting on the body (which is also the constant of proportionality), then

$$y = kx,$$

which is the linear equation representing the given statement.

If $k = 2$ units, the above equation becomes

$$y = 2x.$$

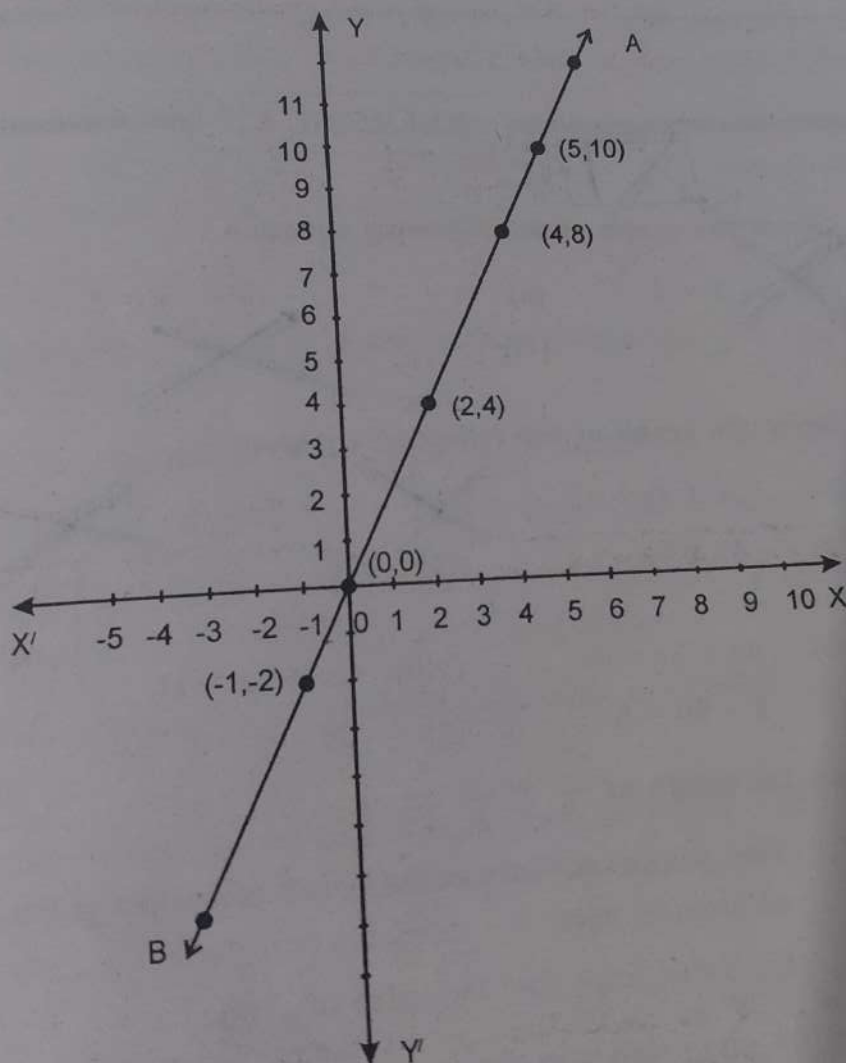


Fig. 4.5

To draw the graph of this equation, we take some of the ordered pairs satisfying the equation as given below :